

Native and non-native avian influence on the soundscape of Tahiti

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Résumé

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1. Introduction

Oceanic island urbanisation and species invasion

Oceanic island are small sized with unique biodiversity and simplified communities shaped by human influence. They stands out as unique ecosystems useful to study the impacts of human activities (Whittaker et al. 2017 ; Russell et Kueffer 2019). Oceanic island tends to be more sensitive to species introduction and extensive anthropogenic land-used modification (Whittaker et al. 2017 ; Spatz et al. 2017 ; Russell et Kueffer 2019). For island in particular, avian species have been among the most concerned taxa, having their community assemblages shaped by both invasion and extensive urbanisation processes (Boyer et Jetz 2014 ; Spatz et al. 2017 ; Rogers et al. 2021).

Interspecific competition and homogenization in islands

Non-native birds are issued from specific families (e.g., Galliniformes, Columbidae, Fringillidae, Passeridae, Estrildidae and Sturnidae) mainly introduced for hunting and aesthetic purposes (Blackburn et Duncan 2001 ; Dyer et al. 2017). These species show peculiar traits adapted to urban and agricultural landscapes, whereas native birds from oceanic island tends to mostly include species occurring in forests (Soares et al. 2021). Non-native species may out-compete native for essential resources, and can affect population dynamic for native species (Davis 2003). As shown with the introduction of the Japanese white eye in Hawaii

(Blackburn et al. 2009). Combined effect of the growing loss of native ecosystems and continued introduction of birds species, especially of forest-dwelling birds, intensify non-native competition threats (Sax et al. 2007). Moreover, the introduction of functionally similar species, promotes negative impact through homogenisation of island bird assemblages (Sobral et al. 2016 ; Sayol et al. 2021 ; Soares et al. 2022).

General context of Soundscape and ecoacoustic, acoustic indices

Advance in soundscape ecology, enabled better understanding of structure and dynamic of the sonic environments (Pijanowski et al. 2011). The soundscape comprehend all components, classified according to the sound sources : geophony (climate and geography), biophony (all living organisms), and anthropophony (human activities) (Pijanowski et al. 2011). Complexity of these acoustic emissions can be characterize via acoustic indices that creates a numeric summary of the soundscape (Farina et Pieretti 2012). Ecological meaning is inferred by relating acoustic indices to diversity metrics (Towsey et al. 2014 ; Eldridge et al. 2018 ; Dröge et al. 2021), habitat conditions (Fuller et al. 2015 ; Sánchez-Giraldo et al. 2021 ; Rudge et al. 2026) or temporal dynamics (Gage et Axel 2014). They can also provide indicators on soundscape disturbances (Villanueva-Rivera et al. 2011).

Implication for oceanic islands

In oceanic islands, ecological complexity is threatened by human pressure through urbanization and invasion. Soundscape study offers a good tool to monitor species invasion, extinction, and assemblage turnover (Farina et Pieretti 2012 ; Ribeiro

et al. 2022). Such examples exist with the study of an invasive ant leading to impoverishment of the soundscape in New Caledonia forests (Gasc et al. 2018). Species assemblages interact acoustically and reflect a dynamic process called an acoustic community (Lellouch et al. 2014; Farina et al. 2016). Acoustic communities are modified by non-native species introduction, with strong implications for native species acoustic habitat and their soundtope quality (Farina 2014; Merchant et al. 2015). The soundtope represents the acoustic pattern created by the distribution of biophony emissions (Farina 2014). Whereas the acoustic habitat encompasses the soundtope and adds the environmental acoustic conditions in which the species assemblage is living (Merchant et al. 2015; Farina et al. 2016).

Acoustic implication

Recent studies have pointed out the potential importance of invasive species as an underappreciated source of novel noise, that could alter acoustic communication among native species (Hopkins et al. 2022) (Hunt et al. 2024). In the context of oceanic island, diminished bird richness reflects fewer vocalizing species, resulting in less saturated soundscapes than mainland (Robert et al. 2019). So, acoustic interference is anticipated to be lessened on islands. Yet, introduction of non-native species is numerous and is expected to continue to increase in islands (Seebens et al. 2017). Compared to native species, loudness and high occurrence rate of vocalization can be characteristic for invasive species (Hopkins et al. 2022). For example, over the course of one year, invasive Pekin Robins (*Leiothrix lutea*) calls made up 37% of all bird songs in a shrubland of their invaded range in Europe (Farina et al. 2013). Likewise, calls of invasive birds such as Common Mynas (*Acridotheres tristis*) seem to dominate the urban environments they have invaded (personal information in (Hopkins et al. 2022)). Plus, timing of calls between native and invasive species are very similar, and may create overlap, important for acoustic competition (Farina et al. 2013; Bleach et al. 2015).

This novel noise is quite attuned to mask vital information encoded in acoustic signals, and may impact frequencies, reduce perception and discrimination (Lohr et al. 2003). This masking effect can negatively affect communication, movement, vigilance, mating and foraging (Shannon et

al. 2016). Affecting individual survival and reproductive success (Halfwerk et al. 2011; Medeiros et al. 2017). Therefore, the calls of non-native species can bring a significant avenue for acoustic competition with native species and may dominate the insular soundscape and favor its homogenization (Farina et al. 2013). Still, evidence in oceanic island is scarce and needs further investigation (Hopkins et al. 2022; Hunt et al. 2024).

Problematisation

Here, we sought to investigate patterns of variation of signal emissions at the acoustic habitat and soundtope level in relation to native/non-native bird assemblage and habitat factors. We predicted that urbanization gradients drive relevant acoustic indices results for the acoustic habitat level (Fairbrass et al. 2017). We also predicted that differences in bird assemblage compel the variation at the soundtope level, and that non-native species in these assemblages have higher implications in acoustic indices computation compared to native species. Given that urbanization gradient and non-native dominated assemblage represent a combined effect on oceanic islands (Soares et al. 2021). We expect that urban sites have more saturated and complex soundscape patterns due to both factors compared to deep forested sites.

2. Methods

Study area

The study was conducted in Tahiti, French Polynesia. Tahiti is the largest (1,042 km²) island of French Polynesia (Monnet et al. 1993). The climate is tropical with an average annual temperature of 26.5 °C and annual rainfall of 1700 mm in the coastal plain of the leeward coast of Tahiti Nui, reaching 4000 mm on the windward coast and Tahiti Iti, and up to 8000 mm in the center of Tahiti Nui (Laurent source qu'est ce qui est pertinent, atlas météorologique quels accès?). Islands of the Pacific Ocean underwent strong impoverishment with numerous bird extinctions due to first Polynesian arrival in the 13th century and Europeans in the 18th century (Thibault 1988) (Bellwood 1978; Monnet et al. 1993; Steadman 1995). From 1960 to 1990, strong habitat modification took place due to development, coastal urbanization and forest clearance (Blanchet 1986) (Monnet et al. 1993). In Tahiti « ... » terrestrial bird species occurs, 12 of them are non-native (Thibault et Cibois 2017). Introduced birds have generally the highest den-

sities in lowland, non-forested habitats such as urban areas or agricultural fields, and lower densities in high-elevation, contiguous forest, where native birds reside (Soares et al. 2021). Similar pattern is observed in Tahiti (Monnet et al. 1993) (besoin source moins loin pour Tahiti or personal observation).

Acoustic data collection

Passive acoustic data was collected using SongMeter SM2 and SM4 digital recording devices (Wildlife Acoustics, Maynard, MA, USA) from 2024 to 2026. In total, 69 sites from different habitat types (Forest = , ... rural or suburban = , .. Urban =) were surveyed in valleys along an altitudinal gradient. Stations recorded along the same valley's gradient have staggering dates, except for « O_Belvedere » where additional recording were conducted. Acoustic recording units (ARUs) were programmed to record continuously and placed at 1,5 meters above the ground. Devices recorded for « ... » per station, with a total of « ... » hours recorded. Recordings were made as .WAV files with a « ... » Hz sampling rate and gain of « ... » dB, with monophonic and 16 Bits formats.

Soundscape index metrics

We selected nine relevant acoustic indices to encompass the sonic emissions heterogeneity for the acoustic habitat and the soundtope of each sampling site. Choice of indices were based on their relevance to reflect patterns of variability, heterogeneity, complexity of both soundscape level. The use of several acoustic indices was prioritized as multi-index analysis have been shown to be more effective than single index investigations (Bradfer-Lawrence et al. 2020). Acoustic indices were estimated for each 10-min recordings. We used indices median output per hour for better control of index variation (Bradfer-Lawrence et al. 2019). Acoustic indices output were normalized to facilitate comparisons.

Two different frequency limits were applied for the upper limit and a low frequency threshold for devices glitches. For the soundtope the limit was (... Hz) to reflect Tahiti bird community emissions. Recordings containing rainfall sounds were excluded from the calculation at this soundscape level. For the acoustic habitat, the limit was (... Hz) to encompass all sonic emissions that could affect the bird assemblages. Recording containing rainfall sounds were included in the calculation

at this soundscape level. All indices were estimated using the default settings with the packages 'soundecology' (...) and seewave (...) packages in R (...).

(Table used for soundscape and bird acoustic community monitoring, with hypothesis)

Habitat metrics

Two metrics, the Hard Surface Area (HSA) and the open herbaceous cover (HERB) were chosen to reflect habitat types and heterogeneity along the altitudinal gradients. HSA and HERB ranges between 0 and 100 %. Respectively, higher values indicates the important cover of human-modified surfaces and open herbaceous areas. Habitat covers map was estimated using polygon surfaces in 100 meter radius circle around each ARUs positions in Google Earth (Google Earth Pro.7.3.7.1327).

Avian characterisation

Two ten minute aural and visual identification in acoustic recordings was conducted for each site, to obtain bird richness count. Identification were conducted from 06:00 am to 06:10 am in recordings free of rainfall or strong acoustic disturbance. From the aural identification database, several assemblage metrics were derived : total richness (count), non-native richness (count), presence of native bird (binomial), presence of Myna (binomial). Non-native bird species encountered were « ... ». Native bird species were « ... ».

Bird assemblage metrics in response to habitat characteristic analysis

Habitat as direct driver of bird assemblage was investigated. Bird assemblage metrics were modeled independently as responses variables using « ... » for or « ... » for « ... » distributions using the habitat metrics and altitude as independent fixed variables. All generalized linear mixed-effects models (GLMER) were made using the « GlmmTMB » package (Brooks et al. 2017).

Acoustic indices in response to bird assemblage and habitat metrics analysis

Prior to analysis, acoustic indices output were tested for correlations.

To investigate if acoustic indices reflect metrics of native/non native bird assemblage, GLMER were used to model each acoustic index (...) individually as response variable, for both acoustic habitat

and soundtope level. In response to the following independent variables : one assemblage metric at a time and habitat metrics. All models included a control fixed effect for the time (hour of the day) to account for temporal autocorrelation and random fixed effects for the site and commune. Models used a « ... » distribution with a « ... » function. All statistical analyses were performed in R v.().

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